Newborn Resuscitation & Postnatal Care Guidelines

HSPTAR MATER DEI

Newborn resuscitation and postnatal care guidelines

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List of abbreviations

A&E	Accident and Emergency (Department)
BE	Base excess
BPD	Bronchopulmonary dysplasia (chronic lung disease of prematurity)
CDH	Congenital diaphragmatic hernia
CPAP	Continuous positive airway pressure
CTG	Cardiotocography
ECG	Electrocardiography
ERC	European Resuscitation Council
ETT	Endotracheal tube
FRC	Functional residual capacity
HR	Heart rate
ILCOR	International Liaison Committee on Resuscitation
IO	Intraosseous
IPPV	Intermittent positive pressure ventilation
LED	Light emitting diode
MDH	Mater Dei Hospital
NPICU	Neonatal Paediatric Intensive Care Unit
PCO_2	Partial pressure carbon dioxide
PEEP	Peak end expiratory pressure
PO_2	Partial pressure oxygen
PPV	Positive pressure ventilation
RCT	Randomised control trial
RDS	Respiratory distress syndrome
SIB	Self-inflating bags
SpO_2	Oxygen saturation
T_i	Inspiratory time
UVC	Umbilical venous catheter

Introduction

Up to 10% of newborns require help to establish breathing but less than 1% require intensive resuscitation[1]. The frequency and level of support required at birth increases with decreasing gestational age at birth.

Newborn resuscitation requires a unique approach and training. Newborns differ from adults not only in size but also in important anatomical and physiological characteristics.

Whilst most collapses in adults are primarily cardiac in origin, the neonatal heart is adapted to the hypoxic experience of childbirth and can withstand up to 20 minutes of severe hypoxia. In newborns collapse is almost always respiratory in origin. Adequate aeration of the collapsed, fluid-filled lungs of a newly delivered baby is usually all that is required for successful resuscitation at birth.

Objectives

The aim of these guidelines is to provide clear and practical instructions on the procedures to be followed in the event of a newborn resuscitation. Such instructions are based on an appraisal of the best available published evidence with adaptations to local logistical considerations.

The remit of these guidelines is the management of the newborn including pre-delivery preparation and post-delivery care but do not include management of the newborn admitted to neonatal intensive care. Separate guidelines exist for management in the intensive care unit.

These guidelines have been developed for all healthcare workers, regardless of their discipline or status, who may be called upon to resuscitate a newborn. These include paediatricians, midwives, NPICU nurses, obstetricians, doctors, nurses working in A&E and family practitioners.

Adherence to these guidelines is essential to provide uniformity of care amongst different health-care providers that is based on the best available evidence and to ensure patient and operator safety.

Newborn resuscitation involves a number of skills which can only be gained through structured teaching and maintained through constant training. These guidelines shall also serve as a framework for continuous training of staff in neonatal resuscitation.

Methodology

These guidelines have been developed following an extensive review of literature published worldwide. Significant weight has been given to the 2010 ILCOR neonatal resuscitation guidelines as these represent an international consensus on the topic by leading resuscitation entities including the European Resuscitation Council, the American Heart Association, Australian together with the New Zealand and South African resuscitation councils.

The latest revision of ILCOR guidance has placed emphasis on the following issues:

- The importance of limiting the amount of supplementary oxygen used during resuscitation
- Monitoring with pulse oximetry from an early stage

- The importance of therapeutic hypothermia for the term or near term babies with suspected hypoxic ischemic encephalopathy
- Delayed cord clamping
- Withdrawal of resuscitation efforts

Where issues have not been covered by ILCOR, guidance has been given based on a systematic review of published evidence including national resuscitation guidelines.

Ethical and practical issues have hindered the piloting of randomised control trials in newborn resuscitation. Where available, the recommendations in these guidelines have been based on RCTs and meta-analyses of RCTs. International opinions tend to diverge on issues which have not been subject to large-scale RCTs. For these issues the guidance provided represents a review of international recommendations. The rationale for any advice given is explained. Local logistical considerations have been taken into account and recommendations have been adapted to the local scene.

1. Delivery room care

1.1 Attendance at deliveries

Prompt initiation is critical to the success of any resuscitation. In most cases the need for resuscitation may be anticipated prior to delivery if risk factors are taken into consideration (Table 1). Unexpected emergencies will still be expected to arise in about 2 out of every 1000 low risk deliveries[2].

Prematurity <37weeks
Antepartum haemorrhage
Mechanical obstruction: Cord prolapse Breech presentation Multiple pregnancy Shoulder dystocia Prolonged labour Precipitate labour Macrosomia
Signs of foetal compromise: Decreased foetal movements on ultrasound Foetal bradycardia on ultrasound Pathological CTG Acidosis on foetal scalp blood sampling
Meconium-stained liquor with evidence of foetal compromise
Caesarean section under general anaesthetic *No increased risk has been demonstrated following planned sections under regional anaesthesia*[3–6].
Anticipated congenital anomalies of the heart, lung and brain
Foetal hydrops
Intrauterine infection or prolonged rupture of membranes > 3weeks

Table 1 – Factors that increase the risk for requiring resuscitation at birth

Every delivery should be assisted by at least 1 person whose primary responsibility is the newborn. This person must be capable of identifying the need for neonatal resuscitation and capable of initiating resuscitation, including administration of positive pressure ventilation and chest compressions. In most deliveries this person will be the midwife.

Deliveries with any of the anticipated risk factors listed in Table 1 should be attended by a clinician with training in newborn resuscitation.

Complicated deliveries, such as those involving extreme preterm babies and babies expected to be born in a poor condition should also be attended by a senior clinician who is proficient in airway management and intubation. Additional help should also be available at short notice should this be required.

1.2 Timing of cord clamping

Up to two thirds of a newborn's circulating blood volume may reside within the placental circulation at birth. Upon delivery, blood continues to flow through the placental circulation for a short period of time resulting in a transfer of blood volume from the placental circulation to the neonate. Such transfer, also referred to as *placental transfusion* may be affected by a number of practices in the active management of labour.

The traditional management of labour has been to clamp and then cut the umbilical cord upon delivery of the baby and prior to delivery of the placenta. This practice has been aimed at reducing the risks of blood loss from both mother and newborn while at the same time allowing access to the newborn should resuscitation be required.

Numerous studies have demonstrated benefit in delayed clamping of the umbilical cord at birth. A meta analysis of fifteen trials[7] has determined that babies who underwent delayed cord clamping had:

- A lesser incidence of anaemia with a decreased requirement for blood transfusions
- A decrease in the incidence of hypotension
- Protection against Grades III-IV intra-ventricular haemorrhage
- Decreased incidence of necrotising enterocolitis[8]
- An improvement in iron status and a number of other haematological indices over the next 3–6 months

The only reported potential adverse outcome was a higher incidence of hyperbilirubinaemia necessitating phototherapy with no increased risk of exchange transfusion[8].

Milking (stripping) of the umbilical cord has also been evaluated and has been shown in one trial to have beneficial effects on haemoglobin and blood pressure with a decreased transfusion requirement similar to delayed cord clamping[9]. A 20cm segment of the cord is milked 2-3 times in the direction of the baby before clamping the cord. This method could provide a rapid alternative to delayed cord clamping in babies likely to require resuscitation.

Delay in umbilical cord clamping for at least 1 minute is recommended for newborn infants, both term and preterm not requiring resuscitation[10,11].

The newborn should ideally be held at the level of the mother's uterus during this delay.

1.3 Temperature control

Newborn babies have large surface area to volume ratios and as such are prone to heat loss and hypothermia. Hypothermia (core temperature < 36.5°C) can lead to surfactant deficiency[12], acidosis, coagulation derangements, delays in transition to a neonatal circulation, hypotension and death[13]. Babies subjected to cold stress immediately after birth have lower oxygen saturations[14] and increased metabolic requirements. Preterm babies are particularly vulnerable to hypothermia given their small size, decreased brown fat deposits (less insulation and energy stores), inability to shiver and immature peripheral vasoconstrictive responses[15] .

Hyperthermia (core temperature >37.5°C) is associated with a poor neuro-developmental outcome and should be avoided at all costs[16,17]. Hyperthermia may occur rapidly, especially in preterm babies placed in plastic bags and resuscitated under radiant heaters.

The optimal temperature in term* and preterm babies is 36.5-37.5°C. This is the temperature range that imposes the least metabolic demand on the newborn[18]**.**

*Unless therapeutic hypothermia is indicated (See section 4.4)

Body heat in newborns can be lost through:

- **Evaporation**: babies are born wet and evaporation of amniotic fluid can cause them to cool rapidly.
- **Convection**: draughts of air around the naked baby and cool air temperatures accelerate cooling. A naked newborn exposed to an environmental temperature of 23°C sustains the same heat loss as a naked adult at 0°C[13]. Babies ventilated or given supplemental oxygen in the delivery room will come into contact with unwarmed and unhumidified hospital gases. Washing babies at birth aggravates convectional heat losses and is unnecessary.
- **Conduction**: babies in contact with cool solid surfaces rapidly lose heat by conduction.
- **Radiation**: newborns lose heat by radiation from exposed skin.

Delivery room and resuscitation room temperature should not be less than 25°C as per WHO guidance[10,19]. This includes operating theatres used for caesarean sections.

Towels should be warmed and the resuscitation surface covered with a warm towel.

Radiant heaters should be switched on during preparations for delivery.

Draughts need to be avoided at all costs. Resuscitation room doors have to be kept closed during ongoing resuscitations.

Babies requiring no resuscitation should be given to their mother for skin-to-skin care (and covered with a towel/blanket) whenever possible.

Term babies and babies born >28 weeks gestation or >1.5kg requiring resuscitation should be dried at birth and kept wrapped in warm, dry towels.

Babies born at < 28 weeks gestation or <1.5kg should be placed wet into a food-grade plastic bag immediately on delivery while still next to the mother[10,11,20–22]. Their head should be covered with a cap.

Babies requiring transfer to NPICU should be transported in a closed transport incubator pre-warmed to a temperature of 33-35°C. The incubator should be kept closed at all times; any handling should take place through incubator port-holes and be kept to a minimum.

Temperature needs to be monitored closely for all babies requiring resuscitation and post-resuscitative transport to NPICU. Skin temperature probes should be applied in the roof of the axilla.

For ventilated babies, warm humidified gases should be used as early as possible.

Newborn babies should not be washed during the first 24 hours.

2. Assessment of the newborn

2.1 Tone

Tone is usually the first sign that can be evaluated at birth. However, assessment of tone is subjective and norms vary with gestational age.

A baby with a flexed posture and moving all four limbs is unlikely to be severely asphyxiated[23].

Babies born limp are more likely to require support. Rapidly proceed to further assessment as detailed below, with particular emphasis on HR and changes in HR.

2.2 Heart rate

- **HR is the most sensitive indicator of a newborn's condition at birth[10].**
- **HR is the most reliable indicator of effective resuscitation. In bradycardic infants HR should respond rapidly to effective resuscitation.**
- **HR in healthy term infants may be <100/min at birth rapidly rising to >100/min within 2 minutes of life[24].**
- **HR < 100/min beyond 2 min of age is abnormal.**

HR should be counted using a stethoscope and not umbilical pulsations.

Count HR over 6 seconds and multiply x 10 for a rapid assessment.

HR should be re-assessed every 30 seconds in babies requiring resuscitation until pulse oximetry is established.

Umbilical pulsations tend to underestimate HR[25] and may result in unnecessary resuscitative efforts.

HR as detected by pulse oximetry is the gold standard as this gives a real-time count and will immediately detect improvement or deterioration.

All infants undergoing resuscitation should be monitored with pulse oximetry[10,26].

2.3 Breathing

Most babies exhibit a strong cry and start breathing regularly within 30 seconds of delivery. This process may take up to 180 seconds in normal term babies[27]. Even in infants born at < 28 weeks gestation, 70% will cry and 80% will breathe spontaneously at delivery[28].

Breathing can be difficult to assess at birth as breathing movements may be subtle. After initial breathing movements the infant may pause for a few seconds and should then establish regular breathing sufficient to maintain a heart rate of >100/min[22].

Newborn babies who do not commence spontaneous breathing or crying within 90 seconds of delivery should be assessed for need of resuscitation.

Infants with HR > 100/min but poor respiratory effort should undergo airway opening manoeuvres and further resuscitation as detailed in section 3.1[10,22,29].

Rate, depth and symmetry of breathing together with any abnormal breathing pattern should be documented[29].

Difficulty in breathing may present after establishment of spontaneous respirations. This may present with the use of accessory muscles of respiration manifested as subcostal, intercostal and sternal recessions and represents increased work of breathing. Expiratory grunting in distressed newborns results from the baby's effort to maintain a FRC by generating PEEP at the larynx.

Tachypnoea, recessions and grunting are important signs of respiratory distress in the newborn. Such babies may benefit from CPAP. This may be initiated in the delivery room using a T-piece device with a facemask as detailed in section 3.2.2(b)[10,22].

2.4 Oxygenation

Newborn babies are born with low arterial oxygen saturations of <25mmHg which increase as the lungs aerate in the minutes following delivery.

- **Colour is a subjective and unreliable sign with significant inter-observer variability[30].**
- **A decision on whether to initiate oxygen therapy and respiratory support should not be taken based on a newborn's colour.**
- **A newborn who is well flexed and moving, with a HR >100/min but is cyanosed does not require oxygen therapy.**
- **It may take up to 10 minutes for healthy term infants to achieve oxygen saturations of 90%.**
- **Pallor is the only sign in the baby's colour which may be of clinical significance and may indicate anaemia, hypovolaemia or acidosis secondary to ante-partum haemorrhage.**

Pulse oximetry is the only standard that should be used to assess oxygenation and should be used for any baby requiring any form or resuscitation.

2.4.1 Pulse oximetry

Pulse oximetry measures oxygen saturation and heart rate by measuring the absorption of light by oxygenated haemoglobin using a LED and sensor.

- **Pulse oximeters give a more accurate count of HR than clinicians using a stethoscope[26] with the added advantage of real-time readings giving an early warning of any change in the baby's condition.**
- **Oxygen saturations can be available within 90 seconds of birth if the probe is placed correctly.**

Pulse oximeters should be attached to the right upper limb at the hand or wrist so as to obtain pre-ductal readings.

Pulse oximeters, like all medical devices are prone to artefact. The following tips will help in rapidly establishing pulse oximetry and obtaining reliable readings:

- Use pre-ductal (right upper limb) readings.
- Applying the oximeter probe to the subject before connecting it to the instrument will produce reliable results more quickly[31].
- Always use appropriate infant probes which are compatible with the monitor.
- LED and sensor need to be properly aligned so as to avoid dispersion through tissues.
- Strong light can interfere with readings. Oximeter probes should be shielded. Limb holders are ideal for this purpose.
- Avoid applying tapes or limb holders tightly around the probe as this will impair perfusion and reduce the oximeter signal[32].
- Movement artefact during transport can be reduced by the use of filtered circuits such as Masimo SET® technology. These are built into GE-Dash 4000® monitors available at MDH.
- Check that the monitor waveform is of good quality.
- If in doubt compare the HR as measured by the pulse oximeter to that measured by ECG electrodes. A difference of >10 beats/min is indicative of a poor trace[33].

2 min	**60%**
3 min	**70%**
4 min	**80%**
5 min	**85%**
10 min	**90%**

Table 2 - Acceptable (P25) pre-ductal oxygen saturations by post-natal age

Oxygen saturations increase gradually from birth, reaching 90% by 10 minutes of age. Babies born preterm and those delivered by caesarean section have lower saturations and exhibit a slower rise when compared to term babies and those born by vaginal delivery[34,35].

Oxygen therapy should never be initiated without saturation monitoring.

Saturation targets set in Table 2 need to be followed rigorously so as to avoid excessive and unnecessary oxygen therapy which may be toxic and impair long-term survival and increase morbidity (see section 3.2.3).

Saturation monitoring should always be used in conjunction with HR monitoring.

Should pulse oximetry not be possible, HR should be checked every 30 seconds by auscultation.

3. Neonatal resuscitation

3.1 Airway management

Most resuscitation scenarios in newborns successfully respond to adequate inflation of the lungs. Collapse in neonates is very rarely a cardiovascular event. Correct airway opening techniques are therefore a prerequisite to successful resuscitation.

Newborns differ from adults in that they have a large head with a prominent occiput as a result of which they tend to flex their neck and collapse their airway when placed supine. The cartilage in the upper airways is softer than in adults making it more difficult for the airway to maintain patency if compressed.

Airway opening manoeuvres constitute the first step in resuscitation once the baby is placed on the resuscitation surface. The baby should be assessed. Tone and respiratory effort/cry may be immediately obvious. A vigorous term baby with a strong cry should be dried, warmed and handed over to his or her mother for skin to skin care whenever possible.

Babies showing poor respiratory efforts may require active resuscitation. Drying the newborn with a warmed towel may help stimulate breathing.

Tactile stimulation whilst drying and warming the newborn with a towel should be carried out for a few seconds.

Heart rate should be assessed and pulse oximetry initiated at an early stage.

The head should be placed in a neutral position; this may be supported by placing a small (2cm) towel roll under the baby's shoulders.

In unresponsive, non-breathing newborns the jaw will need to be brought anteriorly to avoid airway collapse. This can be done using a jaw-thrust manoeuvre by pushing upwards on the angle on the jaw. Care should be taken in any manoeuvre to avoid compressing the soft tissues of the neck[11,22].

Should help be available it may be easier and more effective to apply a double handed jaw thrust.

Any intervention should be followed by re-assessment of the baby with attention to changes in heart rate.

If positive pressure ventilation is required then chest rise should be observed. In the absence of an acceptable chest rise further airway opening manoeuvres should be considered.

The use of oropharyngeal (Guedel®) airways is no longer recommended in preterm babies[10].

3.1.1 Airway suctioning

Foetal lungs are filled with fluid. At birth, part of this fluid is absorbed by the alveolar epithelium. The remaining fluid is expelled from the airways and trachea. It is normal for newborn babies to have an upwelling of fluid from their trachea into their larynx. In vigorous babies this fluid is cleared spontaneously and does not require suctioning even if the baby develops a gargling sound.

Suctioning of the oral and nasal cavities may induce vagally mediated bradycardia, laryngospasm and local trauma[36,37]. Vagally-mediated hypotension may be deleterious, especially in preterm newborns[38]. Blind suctioning of the airways should never be carried out.

Figure 1 - Yankauer suction catheter with bulb head

Airway suctioning should only be considered in non-vigorous babies with a clear suspicion of airway obstruction (such as blood, vernix or meconium).

Suctioning should only be carried out under direct vision using a laryngoscope. If no obvious obstructing matter is visualised then suctioning should be aborted.

Suctioning should be carried out with sterile wide-bore suction catheters (at least 12-14Fr – white and green respectively), preferably of the Yankauer type (Figure 1) as these are suitably wide-bored and have a blunt (bulb-shaped) end which helps reduce trauma.

Suction pressures used should not exceed 100mmHg so as to minimize trauma and suctioning should be kept as brief as possible[11].

3.1.2 Meconium staining of the liquor

The passage of meconium in utero may be a sign of foetal distress and possibly hypoxia. However foetuses pass meconium with increasing frequency with advancing gestational age even in the absence of distress. Aspiration of meconium into the newborn's lungs may cause severe respiratory problems. Meconium is a strong irritant and may cause a severe chemical pneumonitis complicated by pulmonary hypertension. Meconium causes airway obstruction

with a ball-valve effect resulting in patches of collapse alternating with areas of hyperinflation commonly referred to as meconium aspiration syndrome[39,40].

Intrapartum suctioning of the oropharynx has been found to be ineffective[41] as has intubation and suctioning of vigorous newborns born through meconium-stained liquor[42].

Tracheal suctioning of non-vigorous infants born through meconium-stained liquor has not been shown to decrease morbidity and mortality[43,44]. It is thought that in these babies aspiration of meconium would have already occurred in utero as a result of foetal gasping. No randomised trials have been carried out to confirm or refute this practice. NRP guidelines recommend that in the absence of evidence to the contrary this practice should still be considered provided that it does not interfere with other resuscitative efforts.

Airways should never be suctioned in vigorous babies born through meconium-stained liquor.

Suctioning of the larynx and trachea under direct vision for babies born limp through meconium-stained liquor may be considered only provided that this is not prolonged and does not hinder with other resuscitative efforts[21]. A 12-14Fr wide-bore suction catheter should be used.

If the person carrying out resuscitation does not feel confident in carrying out suctioning and possibly intubation under direct vision (laryngoscopy) then this step should be omitted.

3.1.3 Tracheal intubation

Tracheal intubation secures the airway and allows a more controlled delivery of gas into the lungs. It may be considered in the following circumstances:

- Difficult resuscitations where high pressures might need to be used
- Prolonged resuscitation with a need for ongoing ventilation
- Preterm delivery where surfactant administration may be required
- Congenital diaphragmatic hernia
- Upper airway anomalies and the presence of debris in the upper airways
- Stabilisation of the airway for transport

Weight	Miller (blade) size
< 1kg	00
1-3kg	0
>3kg	1

Table 3 - Sizing laryngoscope blades

The oral route is used for intubation during resuscitation. A laryngoscope with a straight blade is necessary as the larynx is more cranially located in newborns than in adults and the use of a curved blade or a blade too long would lead to 'overshooting' the layngopharynx.

Ensure that the laryngoscope blade is of the correct size (Table 3) and that the light is bright with the bulb firmly screwed in place.

Laryngoscope blades, like ETTs should be sterile. Every effort should be made to keep the blade sterile prior and during intubation. Sterile gloves should be used.

Select a correctly sized ETT (Table 4). If using a stylet introducer for intubation ensure that the tip of the introducer does not protrude beyond the ETT tip as this could lead to severe trauma of the airways.

Intubation should be carried out whilst continuously monitoring the baby's HR and saturations using a pulse oximeter.

Each intubation attempt should not exceed 30 seconds.

The most reliable sign that an ETT has been correctly placed is an increase in HR. Other clinical signs are:

- Equality of air entry in both lung fields and the absence of air entry into the stomach.
- Misting of the ETT which may be difficult to see especially in preterm babies.
- Chest excursion, which may be difficult to see and if prominent is usually a sign of over-inflation.
- Exhaled CO_2 detectors allow rapid clinical assessment of ETT placement. These are prone to false negatives in cardiac arrest or in the presence of very stiff lungs.
- The gold standard for checking tube placement is a chest x-ray. This is not usually available at the delivery suite and should be carried out immediately on admission to NPICU. The tip of the ETT should be located at a level between T1 and T4 vertebrae.

Gestational age (weeks)	ETT uncuffed internal diameter (mm)	ETT at lips (cm)
23-24	2.5	5.5
25-26	2.5	6.0
27-29	3.0	6.5
30-32	3.0	7.0
33-34	3.0	7.5
35-37	3.0	8.0
38-40	3.5	8.5
41-43	3.5	9.0

Table 4 - Sizing endotracheal tubes by gestational age[44]

3.2 Respiratory support

At birth the newborn's lungs undergo a transition from a fluid-filled collapsed state to the inflated state such that the lungs maintain volume even on expiration. The consequent rise in the pulmonary volume leads to mechanical relaxation of the pulmonary vasculature. This coupled with oxygenation of the lungs leads to a sharp drop in the pulmonary vascular resistance and to an increase in pulmonary perfusion which allows for effective gaseous exchange.

In an infant born apnoeic this process has not occurred or is incomplete. Any resuscitative efforts should aim to replicate this process. The lungs need to be ventilated in such a way that they maintain a Functional Residual Capacity on expiration. In the majority of cases this will lead to a rise in heart-rate, which is the key indicator of a successful resuscitation.

A significant number of newborns requiring resuscitation will be affected by a deficiency of surfactant, either as a result of prematurity or as a consequence of other pathologies such as meconium aspiration. In these newborns the high surface tension within the alveoli will lead to their collapse on expiration with a resultant increased work of breathing required to recruit the lung on inspiration. Wide fluctuations in pulmonary volume lead to increased shearing forces within the pulmonary interstitial tissues which contribute to lung injury which is reflected in prolonged ventilator dependency and eventual chronic lung disease.

The aim of respiratory support is to assist spontaneous breathing by giving continuous positive airway pressure (CPAP). If the baby is apnoeic, intermittent positive pressure ventilation (IPPV) will need to be delivered over and above CPAP.

CPAP helps with lung aeration, lung fluid clearance and the formation of a FRC. Oxygen exchange is dependent on the surface area available for gaseous exchange and is strongly influenced by the FRC. Maintaining a FRC has a significant benefit on oxygenation resulting in a decrease in oxygen requirements and a decrease in the need for invasive ventilatory support[45,46]. Treating a baby with supplementary oxygen will be ineffective if lung volume is low.

Babies with respiratory distress tend to grunt in order to create PEEP and preserve their FRC in between breaths. Once a baby is intubated the ability to adduct the vocal cords and generate PEEP by grunting is lost. PEEP in these circumstances is completely dependent on what is provided by the resuscitator.

3.2.1 Positive pressure ventilation

PPV is started in those newborns born apnoeic in whom there is no improvement in heart rate following stimulation and airway opening manoeuvres. PPV in the form of CPAP is also applied to spontaneously breathing babies who are in respiratory distress.

(a) Face masks

Round soft silicon facemasks are used in neonatal resuscitation as these fit the newborn's anatomy best and achieve the best seal[47]. Masks are made of transparent or semi-transparent silicon to allow visualisation of secretions. Masks should be big enough to cover the mouth and nose but not too big as to encroach on the eyes.

Both mannequin and live studies have demonstrated that even with appropriate equipment, leaks from around the mask can be significant and can interfere with resuscitation[48]. Correct positioning of the mask can make a significant difference[49].

Select an appropriately-sized mask. Masks should ideally be sterile-packed. If a sterile mask is not available then the mask available should be cleaned thoroughly with chlorohexidine wipes and kept wrapped in a wipe until used.

The mask is applied to the face by first anchoring the lower portion on the chin and rolling the mask upwards. The mask should be held in place using thumb and index finger clamped in a C-shaped configuration around the stem of the mask[50]. Anchoring the mask close to the rims will cause the soft silicone to distort and result in significant leaks. The second and third fingers should be hooked around the mandible to enable the jaw to be pulled upwards (jaw-thrust) and the mask to be firmly applied to the face (Figure 2).

If chest rise is still not seen with this method then the airway should be re-assessed and a two-operator technique should be considered. The requires one operator to apply the mask with both hands and the other operator to deliver PPV[51] (Figure 3).

Figure 2 - Single operator technique

Figure 3 - Two operator technique

Leaks of up to 50% have been recorded with mask PPV in spite of appropriate equipment and technique[48]. In most cases the tidal volume achieved will still be sufficient to stimulate the baby to breathe. In difficult resuscitations endotracheal intubation may be necessary.

3.2.2 Equipment for positive pressure ventilation

(a) Self-inflating bags (e.g. Ambu® bags)

SIBs (Figure 4) consist of a pliable inflatable bag attached to a non-return valve which allows positive pressure inflations to be generated when the bag is squeezed. The outlet can be attached to both facemasks and ETTs. A non-return valve at the patient end and a reservoir bag at the other end allow very high oxygen saturations to be given (>95%) when the SIB is attached to a high flow 100% oxygen source.

Most SIBs are equipped with a spring-loaded blow out valve which is set to open and limit pressures at 40cmH_2O. In practice, blow-out valves open at widely varying pressures and very high pressures (in excess of 70cmH_2O) can be generated with vigorous bagging even if the blow-out valve is not occluded[52].

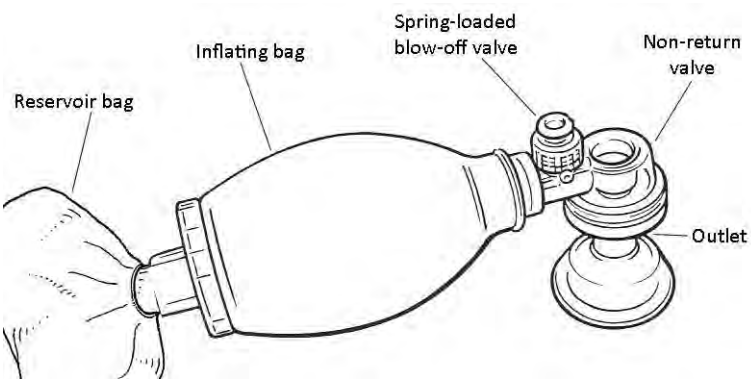


Figure 4 - Self-inflating bag

SIBs are unable to generate CPAP. PEEP valves have occasionally been added to SIBs; however these have been shown to create variable PEEPs and are unsatisfactory[53].

Different volumes of SIBs exist. Traditionally the small 250ml bags have been used for newborn resuscitation. These bags are in the majority of cases unsatisfactory as it is impossible to generate sustained inflations of over 1 second which are required to recruit the neonatal lung at the beginning of resuscitation[54]. 500ml bags are now in common use for neonatal resuscitation; these are squeezed gently between two fingers so as to provide a sustained inflation whilst avoiding excess volume administration.

SIBs have the advantage of not requiring a source of compressed gas to function. Their use should be limited to out of hospital scenarios such as home deliveries and transport where compressed medical gases may not be available. They also constitute an important back-up for in-hospital resuscitation should the gas supply or other equipment fail.

(b) T-piece devices (e.g. Neopuff®)

T-piece systems consist of two units:

- A pressure limiting (wall-mounted) device with manometer which allows the PIP to be set (Figure 5).
- Flexible tuning connected to this device with a T-shaped connection at the end (Figure 6). One limb connects to the facemask or ETT and hence to the patient. At the other

end is an adjustable exhaust valve. Setting the resistance (width) of this valve will adjust the PEEP or CPAP provided to the patient.

T-piece systems require a supply of compressed gas to function but need no electrical supply.

The main advantages of T-piece devices are that they achieve target pressures more reliably than SIB systems (Figure 8) and allow for a PEEP to be provided[50]. The operator has complete control over inspiratory and expiratory times.

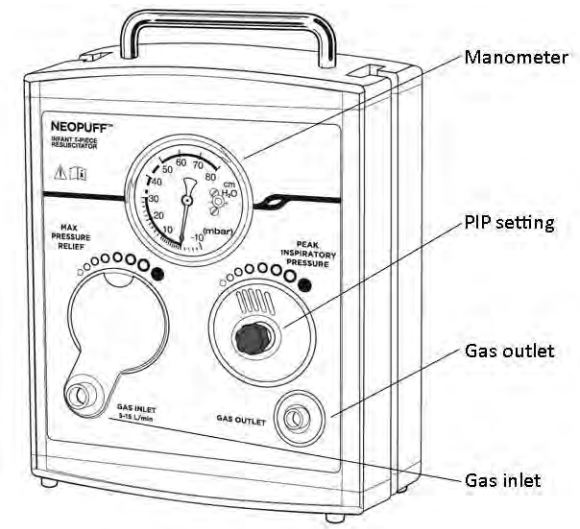


Figure 5 - Neopuff unit

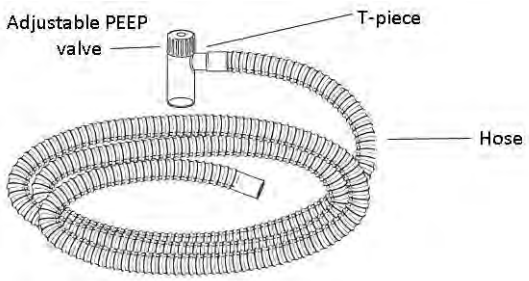


Figure 6 - T-piece with hose

Setting up and using a Neopuff device with T-piece (see Appendix 4):

Ensure that the gas supply is connected to the gas inlet port and open the gas supply to a flow of 8L/min. Do not change this rate of flow during resuscitation as this will upset other settings.

Connect the T-piece with hose to the gas outlet port.

Occlude both limbs of the T-Piece (i.e. the patient end and the adjustable PEEP valve) and use the PIP setting on the Neopuff box to set the PIP, reading the PIP level from the manometer.

Occlude the patient end only leaving the PEEP valve un-occluded. The pressure shown on the manometer is now the PEEP/CPAP. This can be adjusted by turning the adjustable PEEP valve on the T-piece.

The patient end of the T-Piece can now be attached to a facemask or an ETT as required.

Inflations are given by occluding the adjustable PEEP valve with one finger (Figure 7). At this point the pressure in the circuit and therefore in the baby's lungs will not exceed the PIP set on the Neopuff.

Once the adjustable PEEP valve is unblocked the pressure will fall to that set on the PEEP valve, resulting in exhalation. The pressure will not fall to zero. Provided there are no significant leaks it will be sustained at the PEEP level set by the user.

The amount of time spent with the adjustable PEEP valve occluded will be the inspiratory time (T_i).

Figure 7 - T-piece system with PEEP valve occluded

Spontaneously breathing babies who are in respiratory distress can be given facemask CPAP temporarily by applying the mask with T-piece without occluding the adjustable PEEP valve. This will give the baby CPAP at the pressure set on the PEEP valve. The baby will still be able to breathe over and above this pressure.

The flow rate of gas going into the going into the gas inlet should never be varied once the PIP and PEEP are set as doing so could lead to dangerous alterations in pressures[55].

Neopuff with T-Piece	Self-inflating bag
Accurate adjustment of PIP.	PIP varies widely despite blow-off valve.
Allows PEEP to be given and its level to be accurately set.	No PEEP can be given.
T_i can be varied according the duration of PEEP valve occlusion	T_i difficult to vary and is usually very short.
Allows giving facemask CPAP.	Facemask CPAP cannot be given.
Manometer will expose major leaks as set pressures, especially for PEEP will not be achieved.	Leaks difficult to detect
Easier skill to learn[56].	Requires considerable practice, difficult to bag and maintain mask and airway in optimal position at the same time.
Requires a source of compressed gas.	Will still function in room air.
Bulky and difficult to carry around.	Compact, ideal for use in transport and outside hospital.

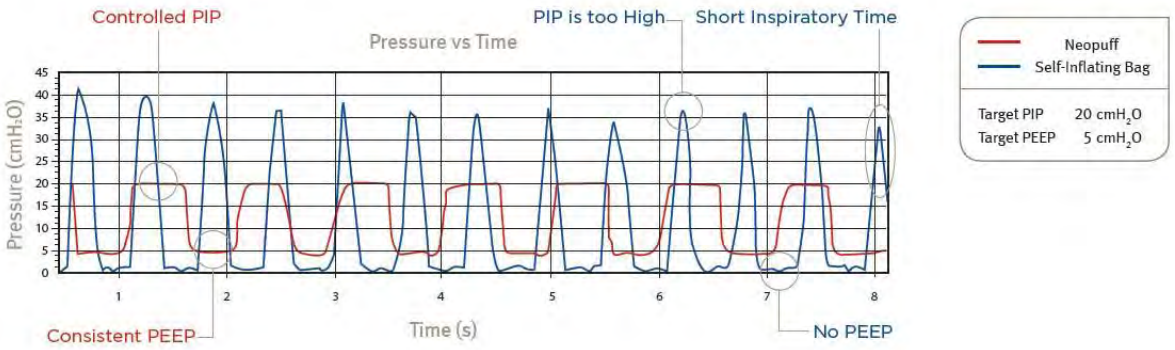


Figure 8 - Pressure performance of SIB compared to T-piece system

3.2.3 Use of supplementary oxygen

Oxygen is probably the most important medication used in neonatal intensive care. Newborns requiring resuscitation have traditionally been treated with 100% oxygen[57]. Randomised studies comparing air vs. 100% oxygen for resuscitation in term and preterm infants have shown that there is a significantly reduced mortality in infants who were initially resuscitated in air[58]. It is thought that high oxygen concentrations lead to the generation of free oxygen radicals which contribute to the establishment of lung disease[59,60] and may also lead to other morbidities such as retinopathy of prematurity[61,62].

Studies in term infants receiving resuscitation with IPPV showed that 100% oxygen conferred no advantage over air in the short term and resulted in increased time to first breath or cry or both[63,64].

Additionally controlled trials have shown that cohorts of preterm babies receiving low concentrations of oxygen (21-30%) had similar oxygen saturations at 5 minutes of age as those who received high concentrations (90-100%)[65].

There is international consensus on the need to use oxygen judiciously.

Oxygen therapy should only be administered under the guidance of continuous pulse oximetry. Oxygen targets by age in minutes should be guided by values given in Table 2.

Oxygen should always be administered with the use of an air/oxygen blender.

Term babies and preterms of >32 weeks gestation requiring resuscitation with PPV should initially be resuscitated in air. If despite effective airway opening and appropriate ventilation there is no satisfactory response in HR, increases in oxygen concentrations may be considered[10].

In preterm babies born at <32 weeks gestation, low initial concentrations of supplementary oxygen are usually necessary. Blended air/oxygen should be started with an oxygen concentration of 30%. If an air/oxygen blender is not available, resuscitation should be started in air[10,11].

(a) Air/Oxygen blenders

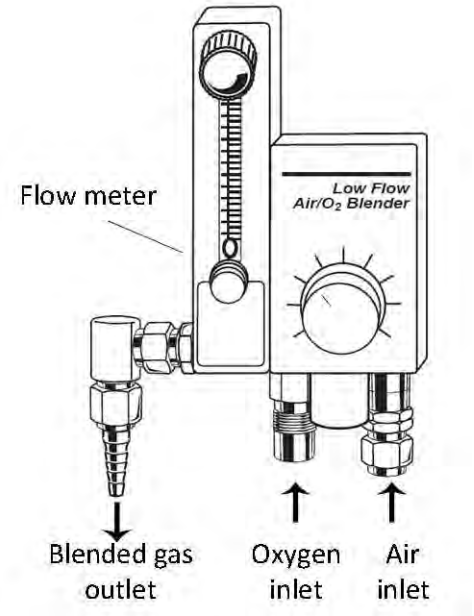


Figure 9 - Air/Oxygen blender with flow meter

Air/oxygen blenders combine compressed air and 100% oxygen from pressurised sources to create a mixture of gas with a desired oxygen concentration ranging from 21%-100%.

Blenders are usually combined with a flow meter (Figure 9) which determines the total flow (l/min) of blended gas being supplied.

The outlet of the blender/flow meter is connected to the inlet port of the Neopuff or SIB.

3.2.4 A strategy for ventilatory management at birth

Tidal volume is proportional to the pressure inflating the lungs and duration of inflation (T_i). In order to achieve a desired tidal volume two possible strategies may be employed: High pressure may be given for a short period of time, however this may be associated with lung injury. Conversely, a lower pressure may be used and sustained over a more prolonged inspiratory time. The second option is likely to be safer and more physiological.

(a) Inspiratory times and rates

Prolonged initial inflation times have been shown in a randomised control trial to increase the FRC[66] and to decrease the need for ventilatory support at 72 hours, shorten duration of ventilation and a decrease in the incidence of bronchopulmonary dyplasia[67].

There is no international consensus on inspiratory times. ERC recommends that:

The first five (inflation) breaths should be maintained for 2-3 seconds.

Subsequent ventilation breaths should be given at a rate of 30-40 breaths/minute.

(b) Inflation pressures

ILCOR and ERC recommend initial PIP values of:

30cm H_2O for term babies, increasing to a maximum of 40cmH_2O if necessary.

20-25cm H_2O for preterm babies.

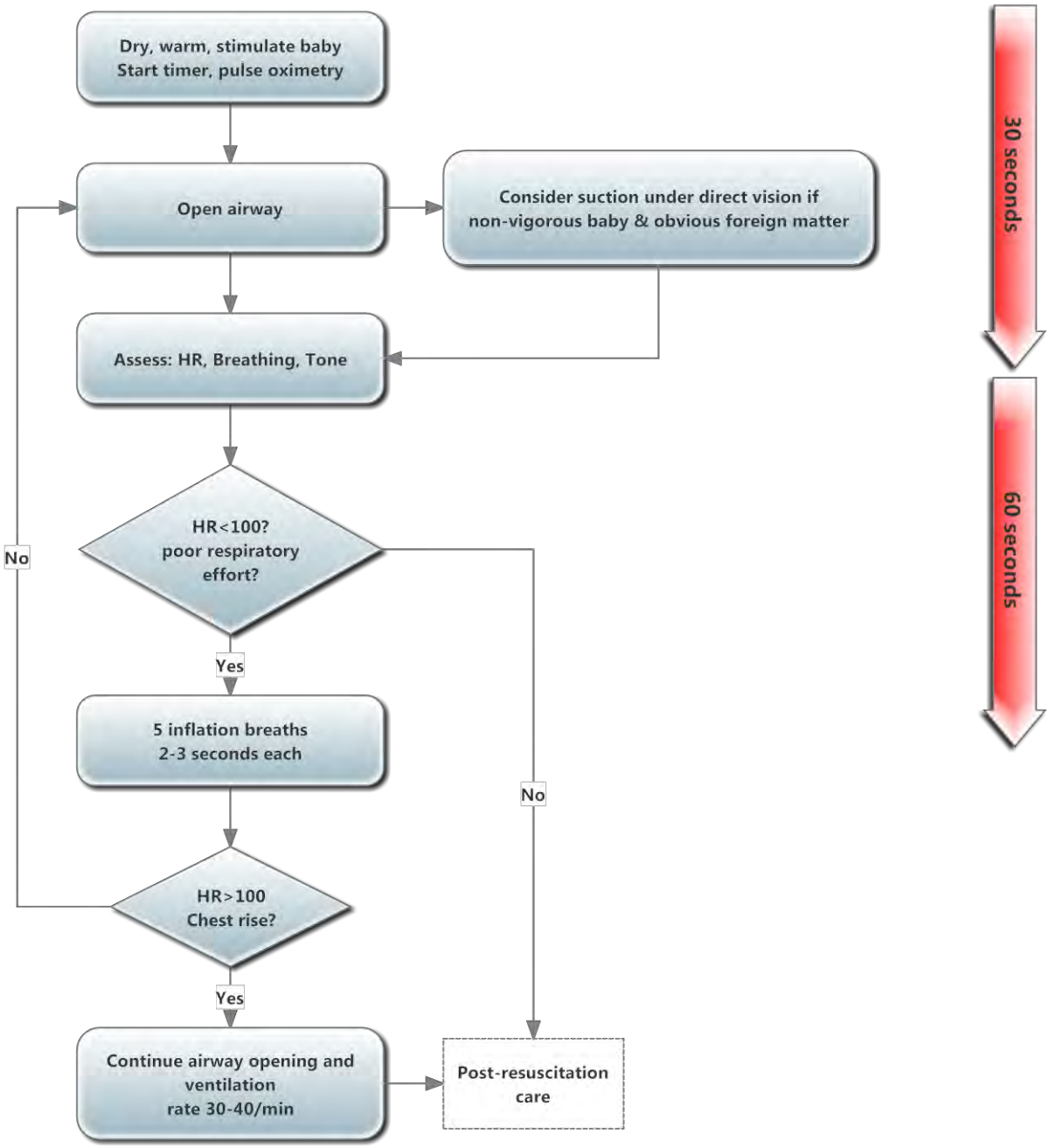
As a general rule, the lowest possible pressure that causes a rise in HR to >100/min should be used.

PEEP should be set at 5cmH_2O for term babies.

Higher initial pressures of up to 8cm H_2O may be necessary in preterm newborns.

Higher PEEPs, especially in term newborns are associated with an increased risk of pneumothorax and should not be used[45].

3.2.5 Summary of respiratory care management



3.3 Circulatory support

Most instances of collapse in the newborn are respiratory in origin. The newborn heart will only slow down or stop if it is subjected to a prolonged period of hypoxia and acidosis. Circulatory support with the use of chest compressions will only be successful if the lungs have been inflated.

3.3.1 Chest compressions

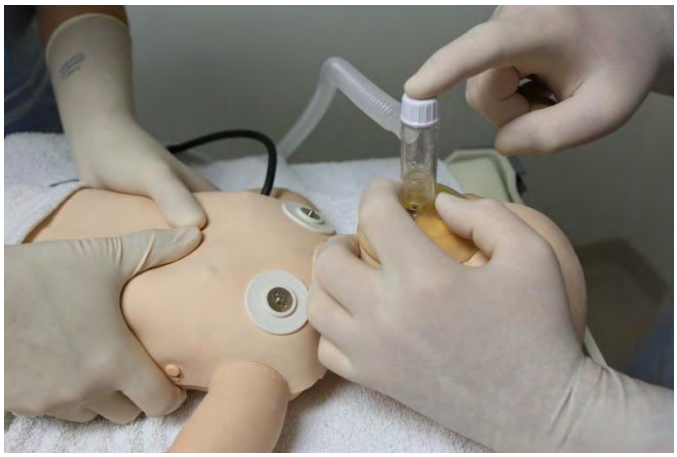


Figure 10 - Chest compression technique

The purpose of chest compressions is to perfuse the myocardium and in doing so increase oxygen delivery and reduce acidosis. Animal studies have shown that the most effective technique for chest compression is the two hand encircling technique as opposed to the single hand two finger technique[66,67].

A higher priority is given to ventilation in neonates. The ratio of 1:3 is based on this principle. However there is no evidence that this is more effective than the 15:2 ratio used in older age groups.

Chest compressions should only be started if the airway has been properly managed, effective ventilation is being provided and the HR is persistently <60/minute.

The chest should be compressed using a two-hand encircling technique with both thumbs pressing on the lower third of the sternum[10] (Figure 10).

The chest should be compressed to 1/3 of the antero-posterior diameter.

Ventilations and compressions should be given at a ratio of 1:3 respectively, at a rate of 30+90 per minute i.e. 120 events per minute[21].

Ventilations and compressions should be co-ordinated so that they do not coincide even if the baby is intubated.

Continuous monitoring with pulse oximetry should be started early.

HR should be re-assessed for response every 30 seconds if oximetry is not available.

3.3.2 Drugs and fluids

Drugs and fluids should only be considered in newborns with severe circulatory compromise (HR <60/min) that have not responded to all other treatment.

The preferred route for administration of drugs in newborns is via an umbilical venous catheter.

All drugs given intravenously or via IO access should be followed by a 1ml 0.9% saline flush.

(a) Adrenaline

Adrenaline is a catecholamine with vasoconstrictive and cardio-stimulatory effects and should be administered via a central vein or intra-osseosly. Only adrenaline at strength of 1:10,000 should be used in resuscitation.

There is no evidence that intra-tracheal adrenaline is effective and safe. Much higher doses need to be used for the intra-tracheal route than are given intravenously. This route should only be used as a last resort if access via all other routes fails[10].

The use of adrenaline in the resuscitation of extreme preterm babies is associated with very poor outcomes (Section 8)

Adrenaline: **10micrograms/kg equivalent to 0.1ml/kg of 1:10,000** for 1st dose

30micrograms/kg equivalent to 0.3ml/kg of 1:10,000 for subsequent doses

50-100micrograms/kg equivalent to 0.5-1ml/kg of 1:10,000 ETT use ONLY

(b) Sodium bicarbonate

Prolonged respiratory and cardiovascular compromise results in severe acidosis. Evidence from animal studies suggests that adrenaline cannot bind to receptor sites in the myocardium in the presence of a low pH.

If there is no response to an initial dose of adrenaline then a dose of sodium bicarbonate may be considered[11].

- Sodium bicarbonate is an alkali solution which is incompatible with catecholamine drugs such as adrenaline. Saline flushes should be used to clear all drugs after administration.
- Only 4.2% sodium bicarbonate should be used for neonatal resuscitation. If only 8.4% solution is available this should be diluted as a 1:1 mixture with water for injection to make up a 4.2% solution.

Sodium bicarbonate: **1-2mmol/kg equivalent to 2-4ml/kg of 4.2% sodium bicarbonate solution**

(c) Volume expansion

Newborns are very rarely volume-depleted at birth. Blood may be lost in the perinatal period through feto-maternal haemorrhage. Such newborns may appear pale and fail to respond to resuscitation. Volume expansion should be considered if foetal blood loss is suspected. Because blood loss may be occult, volume expansion may be considered in newborns who do not respond to resuscitation.

Fluids:	**10ml/kg of 0.9% saline or packed red cells (if available may use O negative blood)**

(d) Glucose

There is no international consensus as to whether glucose boluses should be given during prolonged resuscitation. The UK NLS guidelines suggest that dextrose boluses may be given during prolonged resuscitation if there has been no response to other resuscitative measures including drugs and fluids.

A glucose-containing infusion should be initiated as soon as possible. (See section 5.1 for post-resuscitation glucose homeostasis)

Glucose:	**2.5ml/kg of 10% dextrose**

3.3.3 Vascular access

Newborns requiring drugs and fluids would be expected to have severe circulatory compromise and adrenaline has potent vaso-constrictive properties. Achieving and maintaining peripheral vascular access in these babies is extremely challenging. Drugs administered via peripheral veins would be unlikely to reach the heart in useful quantities. In these circumstances central vascular access is desirable.

Umbilical venous catheters can usually be inserted quickly. Intraosseous access is a viable alternative if intravenous access cannot be established rapidly[10].

(a) Insertion of an umbilical venous catheter

The umbilical vein passes from the umbilicus through the falciform ligament to the left branch of the portal vein where it continues as the ductus venosus to the confluence of the inferior vena cava to the hepatic veins. Both tracts are approximately 2-3cm in length (Figure 11).

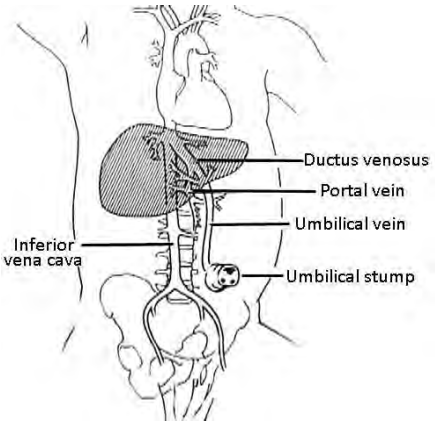


Figure 11 - Anatomical relationships of umbilical vein

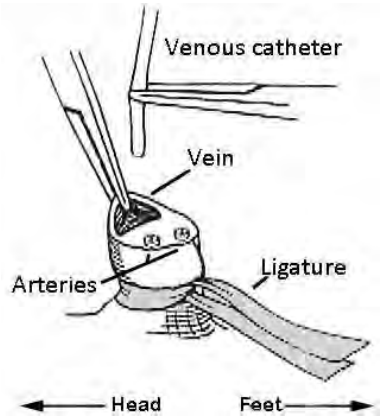


Figure 12 - UVC Insertion technique

The technique for emergency UVC insertion in a resuscitation environment is different from that used in an elective UVC insertion in a controlled environment such as in an intensive care setting.

In emergency vascular access a full aseptic technique may not be possible or be too time consuming. The technique should be as clean as possible with sterile gloves worn and universal precautions taken for your own safety.

Attach a 3-way tap to the UVC and close the tap. This will prevent air embolisation on insertion.

The umbilical cord should be secured with a ligature at the base. This may be tightened in case of bleeding.

The stump should be cut with a clean stroke using a sharp surgical scalpel.

The umbilical vein is usually cranially located to the two umbilical arteries and is usually more patulous in calibre, with a thinner wall.

Support the vein by grasping the edge of the umbilical stump with forceps and gently insert the catheter tip into the lumen (Figure 12).

Advance the catheter into the vein. Some resistance may be felt as the catheter passes the umbilical ring just below the level of the skin. Gentle firm pressure may need to be applied to overcome this resistance.

In an emergency, the UVC should be advanced far enough as to enable blood to be withdrawn.

Once blood is aspirated the length of insertion should be noted from the markings on the catheter. This should be documented. Any bloods required such as blood gas and glucose may be taken at this point.

The catheter is flushed with 2ml of 0.9% saline and fixed to the abdominal wall using an adhesive dressing.

(b) Intraosseous access

Temporary IO access to provide fluids and medications during resuscitation of critically ill neonates may be indicated if attempts to establish umbilical intravenous vascular access are unsuccessful as might be expected in newborns with abdominal wall anomalies such as exomphalos and gastroschisis[10,68].

An 18 Gauge (Pink) IO needle is used in neonates.

The insertion site, 2cm below the tuberosity on the antermoedial surface of the tibia is prepared with an alcohol-based solution.

The limb is immobilised (ensuring that the operator's hand is well clear of that part of the limb to be punctured).

The needle is inserted at 90° to the skin using a firm rotating action. A give may not be necessarily be felt as in older children. Access to the bone marrow will have been achieved at the depth when sampling of the marrow becomes possible.

The IO needle should be flushed with a 0.9% saline solution by connecting it with a three-way tap with extension.

Samples of the marrow may be taken for all urgent blood investigations. The lab needs to be informed of that the sample is marrow and not peripheral blood.

4. Management of newborns at risk of HIE

The prevalence of biochemical asphyxia ranging from mild to severe in newly delivered term infants is reported as being 25 per 1000 live births most of which are classified as mild cases[69]. Even with cord blood gases suggestive of moderate to severe asphyxia most foetuses will not be injured[70].

4.1 Neurological status at birth

The following babies are to be considered as being at **high risk** of developing HIE:

- Apgar scores <5 at 10 minutes of life
- Continued need for resuscitation beyond 10 minutes of life
- Abnormal neurological examination at 10 minutes of life (as detailed in section 4.1.1)

4.1.1 Clinical assessment at birth

Neurological assessment of the newborn is influenced considerably by gestational age[71]. Term newborns who sustain intra-partum insults severe enough to result in long-term sequelae invariably demonstrate clinical evidence of acute encephalopathy during the first days of life[72]. These signs may evolve within the first 4-16 hours and might not be immediately apparent at birth.

In contrast, infants who sustain hypoxic-ischemic cerebral injury earlier in pregnancy may be asymptomatic during the neonatal period only to manifest neurological signs as they grow.

The clinical features of hypoxic-ischemic encephalopathy are nonspecific and similar features may occur in the context of other causes of brain dysfunction.

The level of alertness of a newly born baby in the delivery room does not permit the use of a complete HIE score such as the Thompson score (see section 4.1.3). Such scores should be reserved for later assessments once the baby is settled in the post-natal ward.

The following features should be checked for and documented at birth:

- **Tone**: This is the first sign that can be evaluated and can be rapidly assessed by observing posture and movements. Healthy term babies have a symmetrical flexed posture, most notable at the hips, knees and elbows. Opisthotonic posturing and persistent clenching of the fists is suggestive of hypertonia.
- **Movements**: Normal term newborns move all four limbs in a symmetrical but alternating fashion. Jitters are common but these should be distinguished from seizures in that they do not result in systemic upsets such as apnoea, can be precipitated by stimuli and can be suppressed by passive flexion.
- **Seizures**: Seizures are the most distinctive manifestation of significant CNS disease in the newborn[73].

A newborn baby with a flexed posture who is moving all four limbs is not seriously asphyxiated[23].

4.1.2 Apgar score

The Apgar score was originally created to provide a description of the newborn's condition at birth so as to enable comparison of obstetric and resuscitation practices[74]. It is not a reliable indicator of neurological status at birth and therefore is not a good predictor of neonatal encephalopathy.

- Apgar scores may be affected by non-asphyxial factors such as gestational age, maternal analgesia and infection.
- Apgar scores have a very low predictive value in identifying long-term morbidity[75].
- Severe hypoxic insults may result in a minimal reduction of the Apgar score and most babies with low initial Apgar scores will survive without any neurodisability[75].
- Prolonged depression of the Apgar score is associated with poor survival and neurodevelopmental outcomes[75].
- There is very poor correlation between Apgar scores and cord blood gases[76].

Score	0	1	2
Heart rate	Absent	<100	>100
Respiratory effort	Absent	Weak cry or hypoventilation	Good
Muscle tone	Limp	Some flexion of extremities	Well flexed
Response to stimulation	Nil	Some movement	Cry
Colour	Pale/blue	Body pink, extremities blue	Pink

Table 5 - Apgar score

Of the parameters assessed in an Apgar score the heart rate and respiratory effort are the most important. The other parameters are over-represented (Table 5).

It is customary to record Apgar scores at 1, 5 and 10 minutes of age. Good practice entails providing a full written description of the condition of the baby at birth and response to any interventions.

4.1.3 Thompson score

Thompson scores are a much more reliable tool than Apgar scores in the diagnosis of neonatal encephalopathy (Table 6)[77].

For newborns who have required significant resuscitation at birth it is important to document Thompson scores at 24 and 48 hours.

Grade	0	1	2	3
Tone	Normal	Hyper	Hypo	Flaccid
LOC	Normal	Hyperalert, stare	Lethargic	Comatose
Fits	None	None	< 3 per day	>2 per day
Posture	Normal	Fisting, cycling	Strong distal flexion	Decerebrate
Moro	Normal	Partial	Absent	
Grasp	Normal	Poor	Absent	
Suck	Normal	Poor	Absent ± bites	
Respiration	Normal	Hyperventilation	Brief apnea	Apnoea (IPPV)
Fontanelle	Normal	Full, not tense	Tense	

Table 6 - Thompson score

4.2 Interpretation of cord blood gases

Analysis of umbilical cord blood samples can provide insight to the biochemical status of a newborn both prior to and at the moment of delivery. Comparison of arterial and venous cord blood results may also help provide an aetiology for pathological acidosis.

The interpretation of paired cord blood samples is a complex topic and is dealt with in more detail in Appendix 1.

All normal values for cord gases and lactate refer to arterial samples. It is therefore important for both arterial and venous samples to be available so as to ensure that any decisions are taken on the correct sample.

- The umbilical vein carries oxygenated blood towards the foetus, while the artery removes CO_2 to the placenta. The umbilical vein would therefore be expected to have a slightly higher pH and lower pCO2 than the artery. These parameters should be used to confirm that the samples have been properly labelled.

pH	Pathological foetal acidaemia is defined as an umbilical arterial pH of **<7.0**
Base deficit	Normal arterial base deficit should not exceed **12mmol/L** *i.e. BE_{ecf} less negative than 12mmol/L*
Lactate	Lactate levels **>7mmol/L** are to be considered abnormal

Cord blood gases give a snapshot of the biochemical status of the umbilical cord at the time when the cord is double clamped. Should sudden complete obstruction of the cord occur immediately prior to delivery, cord gases taken at delivery could in fact reflect the biochemical status at the instant prior to the onset of obstruction. Gases taken in such scenarios could therefore be erroneously normal.

A normal cord blood gas is not a substitute for clinical observation. Features of hypoxic-ischaemic encephalopathy should be sought in all at-risk babies (See section 4.1.1).

A low cord pH in an infant who is vigorous at birth and free of cardiopulmonary compromise does not indicate an increased risk of adverse outcome.

- **If in doubt, a repeat blood gas from the newborn should be considered. Infants with a pH of <7.2 persisting at 2hrs of age have a poorer outcome than those in whom acidosis resolved[78] (See Appendix 1).**

(a) When to obtain cord gas samples

- Cord blood gas samples should be obtained in all cases of suspected foetal distress including all emergency caesarean sections carried out for this indication.
- Samples should also be obtained in all other high risk circumstances including severe growth restriction, abnormal foetal heart tracing, maternal thyroid disease, intra-partum fever and multiple pregnancies[79].
- Cord gases should also be obtained in those newborns with Apgar scores <5 at 5 minutes of age as it is possible to do so once the cord has been clamped[80] (see below).

(b) Technique for obtaining umbilical cord samples

In at-risk deliveries, the cord should be clamped in two locations on delivery of the baby as biochemical changes occur immediately[81]. A segment of clamped cord is stable for pH, pCO_2 and pO_2 analysis for at least 60 minutes. Delayed sampling from unclamped cords should never be carried out as results will be misleading.

Only pre-heparinised syringes should be used. Samples in pre-heparinised syringes are stable for pH analysis for up to 60 minutes but should be processed without delay as changes to base excess and lactate levels commence immediately on delivery[81].

Samples should be taken from both an artery and vein whenever possible[80,82]. Ensure that these are appropriately labelled.

4.3 Summary: Management of newborns at risk of HIE

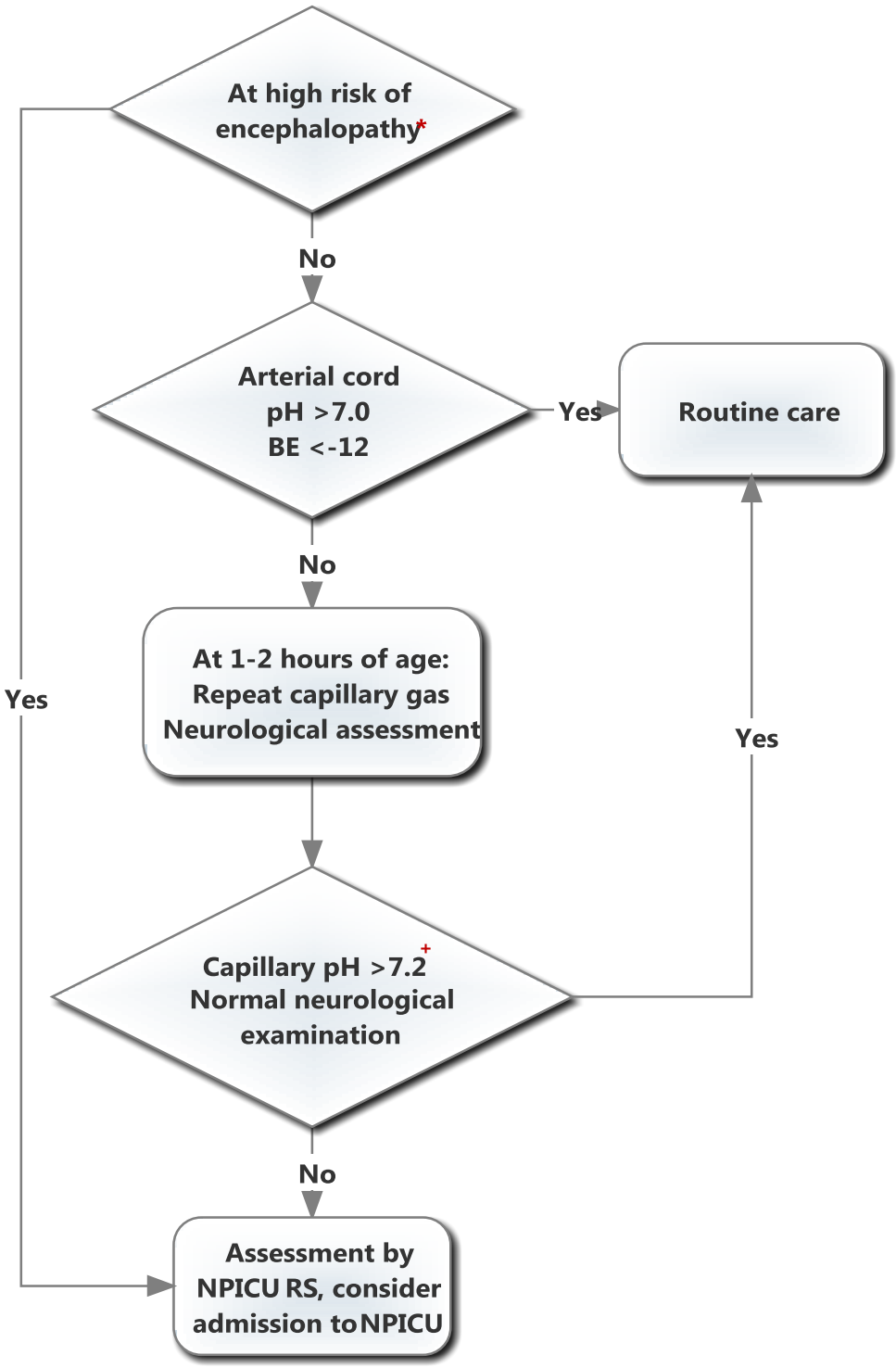


Figure 13 - Summary of cord gas management

* For definition of high risk of encephalopathy see Section 4.1

\+ pH cut-off for pathological acidosis changes from 7.0 to 7.2 at 1 hour of age

4.4 Therapeutic hypothermia

Induced hypothermia in term and near term newborns at high risk of brain injury results in significantly reduced mortality and neurodisability[83–85].

Maximum benefits are attained when cooling is started within 6 hours of birth[86].

For an infant to be considered for therapeutic hypothermia the following criteria need to be satisfied:

A. *Infants **>36** completed weeks gestation admitted to the NICU with at least one of the following:*
   - Apgar score of <5 at 10 minutes after birth
   - Continued need for resuscitation, including endotracheal or mask ventilation, at 10 minutes after birth
   - Acidosis within 60 minutes of birth (defined as any occurrence of umbilical cord, arterial or capillary pH <7.00)
   - Base Deficit 16 mmol/L in umbilical cord or any blood sample (arterial, venous or capillary) within 60 minutes of birth

Infants that meet criteria A will be assessed for whether they meet the neurological abnormality entry criteria B by trained personnel:

B. *Moderate to severe encephalopathy, consisting of altered state of consciousness (lethargy, stupor or coma) AND at least one of the following:*
   - Hypotonia
   - Abnormal reflexes including oculomotor or pupillary abnormalities
   - Absent or weak suck
   - Clinical seizures

The following may be considered for exclusion from therapeutic hypothermia following discussion with a consultant:

Major congenital abnormalities, such as diaphragmatic hernia requiring ventilation, or congenital abnormalities suggestive of chromosomal anomaly or other syndromes that include cerebral dysgenesis.

(Quoted from TOBY protocol[87])

Newborns who meet or are expected to meet the above criteria should be subjected to passive cooling in the delivery room.

Heating devices such as overhead radiant heaters and incubator warmers should be switched off. Incubator ports may need to be left open.

Core temperature needs to be monitored closely in these patients.

NPICU needs to be notified immediately and the baby transferred as soon as stable.

- The target core temperature for therapeutic hypothermia is 33.5°-34.5°C.
- Babies get cold very easily especially in air-conditioned environments. Avoid delays in transferring the baby to NPICU.
- Hyperthermia can be extremely harmful to these babies and should be avoided at all costs.

5. Post-resuscitation care

5.1 Glucose homeostasis

Hypoglycaemia is thought to increase the probability and severity of brain injury in newborns with hypoxic-ischemic encephalopathy[88] however no specific glucose level has been associated with a worsened outcome[89].

Hyperglycaemia has been associated with worsened outcomes in extremely low birth weight newborns[90]. In babies with hypoxic-ischemic encephalopathy hyperglycaemia has not been found to worsen outcome[91] and animal studies suggest that it may be neuroprotective[92,93].

An intravenous glucose infusion should be started as soon as practical after resuscitation[10].

Glucose levels should be checked regularly with the aim of preventing hypoglycaemia.

5.2 Opiates and naloxone

The use of the opiate antagonist naloxone should only be considered in newborns with respiratory depression/distress if there is a documented short-lived maternal exposure to opiates during delivery. Babies most at risk are those whose mothers have been given repeated doses of an opiate 3 hours apart and within 3 hours of birth, especially if these have been given intravenously.

Administration of naloxone to reverse the action of maternal opiates is not part of the resuscitation pathway. HR and oxygenation should be established and maintained by adequate ventilation.

- **Naloxone is not a resuscitation drug. Administration of naloxone should never be allowed to interfere with the neonatal resuscitation pathway.**
- **Naloxone should never be given to newborns of mothers with chronic opiate use as this may precipitate seizures.**

Naloxone: **100micrograms/kg equivalent to 0.25ml/kg of 400micrograms/ml given im or iv**
(iv doses work faster than im but have a shorter duration of action)

5.3 Surfactant therapy

Although not strictly part of neonatal resuscitation, prophylactic surfactant therapy is frequently administered in the delivery suite on intubation of newborns with respiratory distress.

Endogenous pulmonary surfactant reduces surface tension at the air-liquid interface of the alveoli during ventilation and stabilizes the alveoli against collapse. A deficiency of pulmonary surfactant in both preterm and term newborns results in respiratory distress.

The prophylactic use of surfactant results in a significant reduction in mortality, pneumothorax and BPD in preterm infants[94].

Two formulations of natural surfactant are available at MDH. Poractant alpha (Curosurf®) is a porcine lung extract while Beractant (Survanta®) is a bovine lung extract. Survanta® contains 25mg/ml of phospholipid while Curosurf® contains 80mg/ml (Table 7).

Both manufacturers advise that surfactant be administered through a small-bore catheter inserted into the endotracheal tube. Early instillation of exogenous surfactant appears to reduce the lung injury and protein leak associated with preterm delivery[98]. This protection is diminished if therapy is delayed beyond 15 minutes of birth[99] hence the importance of administering the first dose in the delivery suite.

For prophylaxis of respiratory distress Survanta® is only licensed for use as a 100mg/kg dose while Curosurf® can be administered at a dose raging from 100-200mg/kg.

A meta-analysis of studies comparing the two has found that in preterm newborns of <32 weeks gestation a prophylactic dose of 200mg/kg of poractant results in a more significant decrease in oxygen requirement, allowing faster weaning of oxygen and ventilation, when compared to a 100mg/kg dose of beractant[95]. One study has also suggested that there may be a decrease in mortality with a 200mg/kg prophylactic dose of poractant compared to a 100mg/kg dose of beractant[96], while another study has also hinted at a reduction in the incidence of pneumothorax and intracranial haemorrhage[97].

When administering poractant prophylaxis to preterm newborns it would therefore be advisable to round up doses to as close to 200mg/kg as possible and avoid giving doses of less than 100mg/kg.

Generic name	Brand name	Source	Phospholipid content	Recommended dose for prophylaxis	
			mg/ml	mg/kg	ml/kg
Poractant alpha	Curosurf®	Porcine lung	80	100-200	1.25-2.5
Beractant	Survanta®	Bovine lung	25	100	4

Table 7 - Comparison of surfactant preparations available at MDH[100,101]

Prophylactic surfactant should be administered in the delivery suite as soon as the newborn is stabilised and intubated.

Surfactant should be allowed to warm at room temperature for 15 minutes prior to administration. Each phial should be inspected for particulate matter and discoloration. The solution should not be shaken.

ETT tip position should be ascertained clinically prior to instillation of surfactant by confirming that the length of insertion marked on the ETT agrees with the length appropriate for the size of the child (Appendix 3). Equality of air entry should be auscultated for on both sides. The head should be in the midline position during administration. It must be ensured that the catheter tip does not protrude beyond the ETT tip.

HR and SpO_2 should constantly be monitored during the procedure.

5.3.1 When to give surfactant [102,103]

- Available evidence suggests that exogenous surfactant administration is associated with a highly significant reduction in mortality and air leaks in infants born at a gestational age of 25-29 weeks. Surfactant therapy should thus be considered in all infants born within this gestational age range who are intubated for respiratory distress following delivery.
- There are numerous studies that demonstrate a significant clinical improvement in infants born at 29-32 weeks. Newborns born at 29-32 weeks gestation who have been intubated for respiratory distress should receive surfactant therapy.
- There is very little randomised controlled trial evidence on infants born >32 weeks. However, it would seem appropriate to consider surfactant replacement therapy in an infant of a gestational age of 32 weeks or more who has been ventilated for respiratory distress and where surfactant deficiency is felt to be playing a significant role in the respiratory disease. Use of exogenous surfactant in these newborns should first be discussed with a consultant.

6. Special situations

6.1 Tension pneumothorax

Minor air leaks occur frequently even in term newborns who require no resuscitation and are mostly asymptomatic[104]. The presence of pulmonary disease such as RDS, pulmonary hypoplasia and MAS predisposes to pneumothorax. Iatrogenically induced pneumothoraces may be the result of vigorous resuscitative efforts with high inflation pressures and the use of high PEEPs especially in term babies who are able to generate high pressures over and above the given PEEP[105].

Large pneumothoraces result in respiratory embarrassment from compression of lung tissue and cardiovascular compromise from compression of the superior and inferior vena cava.

Tension pneumothorax predisposes to intraventricular haemorrhage in preterm newborns[106].

Suspect tension pneumothorax in any newborn with a sudden deterioration in respiratory or cardiovascular status or failure to respond to resuscitation.

- Physical signs of asymmetric chest movement and unequal air entry may have to be relied upon.
- Transillumination of the chest may be helpful in preterm neonates but is rarely of any value in term babies.
- Chest x-ray is the gold-standard in diagnosing pneumothorax, however there will not be sufficient time in the delivery room for this to be carried out if the baby's condition is critical.

Emergency decompression of a tension pneumothorax should be performed in the delivery suite if the baby is critically unwell. All other pneumothoraces may be drained in NPICU following a chest x-ray.

A 21G (green) butterfly needle should be connected to a three-way tap which is held under water. Avoid aspirating with a syringe as this might result in excessive drainage of air which will rapidly collapse the intrapleural space causing the needle to perforate the lung. A tension pneumothorax should by definition be under pressure and the air should bubble out by itself.

Insert the needle at the second intercostal space in the mid-clavicular line until air bubbles from the three way tap. Avoid excessive insertion to avoid trauma to the lung.

All emergency decompressions should be followed up by the insertion of a definitive chest drain and a chest x-ray.

6.2 Congenital anomalies

Many major congenital anomalies are diagnosed by antenatal ultrasound and this should allow time for preparation. Some anomalies might still present unexpectedly at birth. Proper awareness of these potential problems is necessary to enable appropriate diagnosis and management.

The presence of senior medical staff during delivery of newborns with suspected major or life-threatening congenital anomalies is of utmost importance. NPICU staff and all health care providers who are envisaged to be involved in the management of the newborn should be notified in advance.

6.2.1 Congenital diaphragmatic hernia

If not antenatally diagnosed this condition should be suspected in the newborn with respiratory failure that does not respond to facemask PPV. A prominent chest with a scaphoid abdomen may be a feature but is usually difficult to discern during resuscitation.

If CDH is anticipated, the baby should not receive facemask PPV as this could inflate intra-thoracic bowel and aggravate splinting of an already hypoplastic lung.

A newborn with known CDH and respiratory compromise should be intubated and given PPV at birth.

A wide-bore naso-gastric tube should be inserted soon after intubation so as to decompress any intra-thoracic bowel.

Newborns with CDH frequently have hypoplastic lungs which are at risk of pneumothorax. Pneumothorax should be strongly suspected if there is clinical deterioration or a poor response to resuscitation.

6.2.2 Upper airway anomalies

These almost always occur unexpectedly as they are difficult to diagnose antenatally.

Suspect choanal atresia in babies who are vigorous at birth and who become pink when crying but profoundly cyanosed when quiet.

Newborns with micrognathia may also have a posteriorly displaced tongue which may obstruct the airway (glossoptosis).

Most upper airway obstructions can be temporarily bypassed with an orophyaryngeal (Guedel) airway. Intubation may be difficult.

Newborns with glossoptosis may benefit from being placed prone.

6.2.3 Exomphalos and gastroschisis

Major abdominal wall defects are usually diagnosed antenatally.

The abdomen should be covered with cling film or a food grade plastic bag to minimise heat and fluid losses pending urgent surgical review in NPICU.

7. Neonatal transport

Neonatal transport refers to the transfer of a newborn baby from one safe environment to another. The same principles apply for transfer between wards within the same hospital as to between units in different hospitals. Each transfer needs to follow a structured approach:

1. **Control**: A team leader needs to be designated to take on overall responsibility for the transfer. This is usually the most senior clinician present on site. The team leader will assign specific tasks to other team members.
2. **Communication**: The team leader is the person who must ensure that all communication occurs appropriately. He or she may however delegate to a competent member of staff. Communication should include:
    a. Consultant on call, who needs to be informed of all transfers
    b. Medical staff at the receiving unit
    c. Nursing staff
    d. Paramedics e.g. ambulance crew, porters (if required)
    e. Parents/legal guardians
3. **Pre-transport assessment**: using a structured ABCDE approach to ensure that the patient is stable and fit for transport and that stability is maintained throughout the transfer:

Airway	The child must have a stable airway, if any doubts exist about the stability of the airway, then elective intubation must be considered. Suction must be available. ETT position must be checked clinically and the length of insertion checked and documented. ETTs need to be appropriately fixed to avoid displacement. Ensure that a nasogastric tube has been inserted. Check and document position. End-tidal CO_2 monitoring should be considered for all inter-hospital transfers as this provides an early warning of ETT blockage/displacement.
Breathing	Bilateral air entry should be ascertained. All transfers should occur with pulse oximetry preferably using monitors that filter out movement artefact. Babies who are intubated should be ventilated with a portable ventilator or t-piece system that can deliver reliable pressures and provide PEEP. If the baby is ventilated, portable ventilators need to be checked prior to use. Any chest drains need to be properly fixed and connected to a flutter valve as underwater seals are inappropriate for transport. The depth of insertion of the chest drain must be documented. For long transfers blood gases may need to be checked en route.

Circulation	HR should be monitored with pulse oximetry. For inter-hospital transfers ECG monitoring is mandatory. Vascular access must be properly fixed and documented including umbilical line insertion depth. BP monitoring should be available for long transfers, preferably invasive.
Disability	The baby's level of alertness should be assessed and documented prior to transfer. Should a prolonged transfer be required, consideration should be given to analgesia and even paralysis in the invasively ventilated neonate. Newborns rapidly become hypoglycaemic. A glucose infusion should be started during all inter-hospital transfers.
Exposure	Newborns rapidly become hypothermic. Transfer incubators should always be pre-warmed. Warm towels may need to be placed around the newborn to maintain temperature with care to avoid hyperthermia. For long transfers consideration should be given to humidifying the incubator.

4. Documentation: the following should be included:

- Baby's details
- Timings: date/time of contact, transport and critical events
- Clinical baseline history and examination
- Clinical interventions and their effects
- Investigations carried out e.g. gases and their outcomes
- Condition during transfer
- Names of responsible clinicians at each stage of transfer
- Parent details and contact numbers (if transferring outside hospital)
- Signature, medical council number and designation of recording physician

7.1 Logistical considerations

See Appendix 6 for a full list of equipment.

- Battery supply: Batteries should be continuously kept on charge when not in use
- Gas supply: Always ensure that gas cylinders are completely filled prior to use

Calculating air and oxygen supply:

D-sized gas cylinders used on transport are usually filled to 137bar and contain **340L** of compressed gas.

If a gas flow-meter is set at 10L/min a full D-sized cylinder will be empty in 34 minutes. Double the calculated required volume should be carried as backup.

Gas driven ventilators such as those used for transport will consume approximately 1L/min of oxygen to drive the ventilator over and above the tidal volume requirements of the baby. Such consumption will need to be factored into any calculation of gas requirements.

8. Ethical issues and communication with parents

8.1 Non-initiation of resuscitation

Whenever the delivery of a neonate at the margins of viability or with conditions which predict a high morbidity and mortality is foreseen effective communication with parents should take place without delay. This should be led by the most senior clinician, preferably a consultant. The views of the parents regarding resuscitation should be supported whenever possible.

ILCOR consensus statement recommends that:

- In conditions associated with a high rate of survival and acceptable morbidity, resuscitation is nearly always indicated.
- When gestation, birth weight, or congenital anomalies are associated with almost certain early death and an unacceptably high morbidity is likely among the rare survivors, resuscitation is not indicated. The British Association of Perinatal Medicine has published the following recommendations about resuscitation of extremely preterm infants[107]:

< 23 weeks:	It is in the best interests of the baby for resuscitation not to be carried out as births at this gestation are associated with very poor survival rates and a high risk of disability[108].
23^{+0}-23^{+6} weeks:	Resuscitation may only be started if a foetal heart has been detected in labour and parents have expressed an opinion in favour after appropriate counselling. A decision on whether to proceed with resuscitation will depend on the response in HR to lung inflation by facemask.
24^{+0}-24^{+6} weeks:	Resuscitation should be started unless the parents have requested for it not to start after appropriate counselling. If the newborn appears to be more immature than expected, a decision not to initiate resuscitation may be appropriate. A decision on continuing resuscitation will depend on the response in HR to lung inflation by facemask.
>25 weeks:	Resuscitation should be initiated chances of survival are significantly better[108].
If gestation is uncertain, a decision on whether to initiate resuscitation should be taken at birth, based on the maturity of the newborn and the condition at birth. A decision on continuing resuscitation will depend on the response in HR to lung inflation by facemask.	
Always double check gestational age yourself using both LMP and dates by ultrasound	

Resuscitation drugs in newborns born <26 weeks gestation

The use of adrenaline in neonatal resuscitation is associated with a poor outcome even if the infant survives the initial collapse[109] and in very preterm infants this outcome seems to be even worse[110]. The British Association of Perinatal Medicine advises that in planning resuscitation of premature infants, especially those born before 26 weeks gestation, a prior decision may have to be taken in conjunction with the parents on whether or not to offer drugs and adrenaline[111].

8.2 Discontinuation of resuscitation

Little good quality evidence exists as a guide to the discontinuation of resuscitation. Published studies show that babies born without a heart rate and who remain asystolic at 10 minutes of age are likely to die or have severe neurodisability[112,113]. No reliable evidence exists for those newborns who remain persistently bradycardic (HR<60) but have a detectable pulse.

ILCOR consensus statement recommends that provided no reversible causes are identified:

- In a newly born baby with no detectable heart rate that remains undetectable for 10 minutes, it is appropriate to consider stopping resuscitation.
- The decision may be influenced by the parents' previously expressed feelings about acceptable risk of morbidity.

8.3 Communication with parents

The birth of a new baby is a very important event for both parents and anxiety will be high, especially if there has been any form of resuscitation.

Information given to parents must be factual and non-judgemental of any other care they may have already been given.

Anxiety interferes with the retention of information. Technical terms should be avoided and only essential but honest information should be given initially.

Whenever possible, parents should be allowed to touch and even hold their baby.

Appendix 1: Interpreting cord blood samples

All values mentioned in this section refer to arterial samples unless otherwise specified.

pH

Pathological foetal acidaemia is defined as an umbilical arterial pH of **<7.0**. It is unlikely that acute acidosis of lesser severity could be directly associated with adverse neurodevelopmental outcomes[114].

It is important to distinguish between respiratory and mixed respiratory/metabolic acidosis. Isolated respiratory acidosis usually results from short-lived impairment in foetoplacental circulation and does not usually lead to severe outcomes. Prolonged foetal acidaemia due to persistent hypoxia results in a mixed respiratory and metabolic acidosis and carries a worse outcome.

pCO_2

Cord compression results in CO_2 retention, similar to airway obstruction in children and adults. Moderate compression leads to a respiratory acidosis with a low pH, normal base deficit and a high pCO_2. More severe compression will cause a combined respiratory and metabolic acidosis with a low pH, high pCO_2 and a high (more negative) base deficit.

Base deficit

No adverse neonatal outcomes have been noted with an <u>arterial</u> base deficits <12mmol/L. Moderate to severe complications occur in 10% of newborns with a base deficit of 12 - 16mmol/L and in 40% of newborns with a base >16mmol/L[69,114].

pH is a logarithmic measurement; therefore the fall in pH does not give a proportionate indication as to the severity of acidosis. Base deficit is a linear measurement. The deterioration in base excess provides a more reliable indicator of the severity of metabolic acidosis and is therefore of greater clinical significance than pH which gives no indication as to the origin of acidosis.

Note that the blood gas machines issue base excess. This value is calculated by the machine using pH and pCO_2 values. The base excess for extracellular fluid (standard base excess) BE_{ecf} should be used as this provides a better representation of the true acid-base status of tissues than the actual base excess (base excess whole blood - BE_B).

Lactate

Foetal lactic acid production indicates a switch to anaerobic metabolism as a result of intra-uterine oxygen insufficiency. Lactic acid crosses the placenta in very small quantities and therefore any measured lactate is mostly foetal in origin. This can provide an indication of asphyxia that correlates well with pH and base excess[115].

Lactic acid is measured directly by the blood gas machine making it more reliable parameter than base excess which is calculated by the machine and may be adversely influenced by spurious pCO_2 values[116].

No international consensus exists as to what constitutes an abnormally high <u>arterial</u> cord lactate level. It has been shown that lactate levels vary by gestational age and mode of delivery[117]. **It would be prudent to consider any lactate of >7 as being abnormal**.

Serum lactate levels in the first 30 minutes of life do not predict the severity of encephalopathy. In contrast, raised serum lactate at 30 minutes of age predicts the development of hypoxic–ischaemic encephalopathy and the time to normalisation of lactate levels is a reliable predictor as to the degree of encephalopathy and severity of seizures[118]. It would therefore be prudent to re-test newborns with abnormally high levels of lactic acid after 30-60 minutes.

Understanding paired arterial and venous cord gas samples

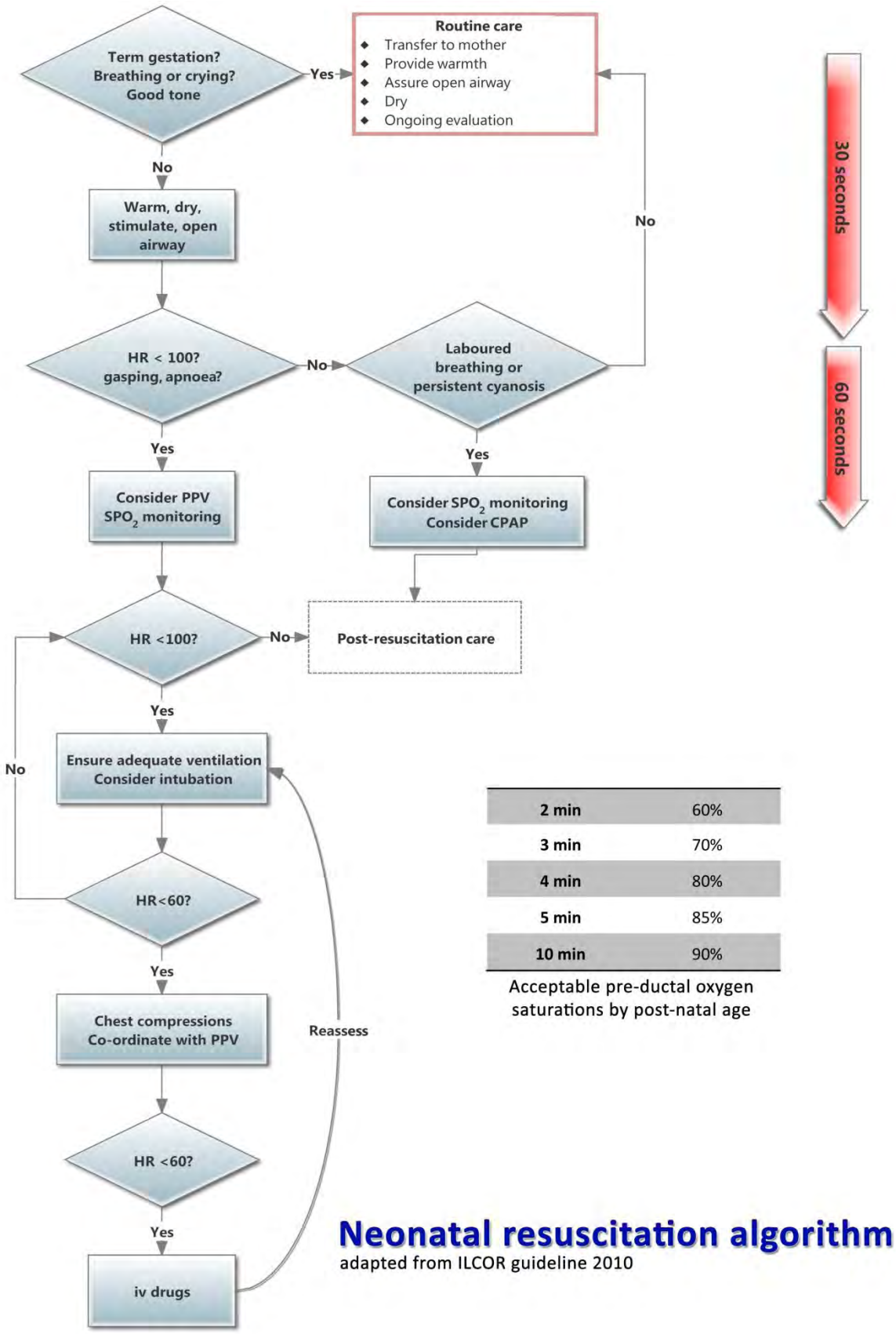
Paired arterial and venous cord samples are frequently taken in the delivery suite. Comparison of arterial and venous cord gases can provide insight to the aetiology of acidosis.

The following points should be considered when interpreting paired samples [119]:

- The gap between arterial and venous gases opens up when there is obstruction to umbilical blood flow. This can be used as a marker of antenatal cord obstruction e.g. cord prolapse.
  - The difference in veno-arterial pH should not exceed 0.15 units. It may exceed 0.3 units in severe asphyxia[120].
- The gap between arterial and venous gases does not widen when there is a restriction to perfusion on the maternal side of the placenta as would occur in placental abruption.
- The magnitude of the difference between arterial and venous pCO_2 predicts the risk of the infant developing encephalopathy in cases of suspected asphyxia (i.e. if the arterial pH is <7[121]).
  - The arterio-venous pCO_2 difference should not exceed 15mmHg.

Results should be interpreted with caution. It is very easy for the wrong vessel to be sampled. If arterial and venous gases are identical then assume that both samples have originated from the same vessel, usually the vein, as this is the easiest to sample.

Appendix 2: Neonatal resuscitation algorithm



2 min	60%
3 min	70%
4 min	80%
5 min	85%
10 min	90%

Acceptable pre-ductal oxygen saturations by post-natal age

Delivery room resuscitation drugs and intubation

Gestational age (completed weeks)	Estimated weight[1] (kg)	ETT uncuffed internal diameter[2] (mm)	ETT at lips (cm)	Laryngoscope blade size	Surfactant prohylaxis (1st dose – within 15min of life) Curosurf® 80mg/ml (mL)	Surfactant prohylaxis (1st dose – within 15min of life) Survanta® 25mg/ml (mL)	Intravenous Fluids♦ 0.9% saline or O neg blood (mL)	Intravenous Adrenaline♦ 1 in 10,000 (mL) 1st dose	Intravenous Adrenaline♦ 1 in 10,000 (mL) Subsequent doses	Intravenous Sodium bicarbonate 4.2%♦ (mL)
					1.25-2.5mL/kg ETT	4ml/kg ETT	10mL/kg iv	0.1mL/kg iv	0.3mL/kg iv	2mL/kg iv
23-24	0.5	2.5	5.5	00	1.25	2	5	0.1*	0.3*	1*
25-26	0.8	2.5	6.0		1.5	3	8	0.1*	0.3*	2*
27-29	1.2	3.0	6.5	0	3.0	5	10	0.1	0.3	2
30-32	1.7	3.0	7.0		3.0	7	15	0.1	0.3	3
33-34	2.2	3.0	7.5		3.0+	9+	20	0.2	0.6	4
35-37	2.7	3.0	8.0		4.5+	11+	25	0.3	0.9	5
38-40	3.2	3.5	8.5	1	4.5+	13+	30	0.3	0.9	6
41-43	3.5	3.5	9.0		4.5+	14+	35	0.4	1.2	7

♦The preferred route for intravenous drugs is via umbilical venous catheter. All drugs given intravenously or via IO access should be followed by a 1mL 0.9% saline flush.
*Administration of adrenaline and other resuscitation drugs in newborns born <26 weeks gestation is associated with very poor outcomes[3].
+Administration of exogenous surfactant to newborns born >32weeks gestation should occur after discussion with a consultant.

Glucose
Dextrose boluses may be considered during prolonged resuscitation if there has been no response to other resuscitative measures including drugs and fluids.
Intravenous glucose infusion should be started as soon as practical after resuscitation with the aim of preventing hypoglycaemia

Naloxone
The use of naloxone should only be considered in newborns with respiratory depression/distress if there is documented short-lived maternal exposure to opiates during delivery.
Naloxone should never be given to newborns of mothers with chronic opiate use as this may precipitate seizures in the newborn.

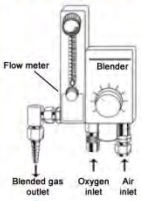
Estimated weight (kg)	10% dextrose (mL)	Naloxone 400micrograms/mL (mL)	Morphine 10mg/mL (mL)	Atropine 600micrograms/mL (mL) §	Suxamethonium 50mg/mL (mL) §
Dose/kg	2.5mL/kg iv	100micrograms/kg im	0.1mg/kg iv	20micrograms/kg iv	2mg/kg iv
1	2.5	0.25	1.0	0.03	0.04
1.5	4.0	0.40	1.5	0.05	0.06
2	5.0	0.50	2.0	0.06	0.08
2.5	6.5	0.65	2.5	0.08	0.10
3	7.5	0.75	3.0	0.10	0.12
3.5	9.0	0.90	3.5	0.12	0.14
4	10.0	1.00	4.0	0.13	0.16
4.5	11.5	1.00	4.5	0.15	0.18
5	12.5	1.00	5.0	0.17	0.20

§For small drug doses use appropriate 'tuberculin' type syringes for accurate dosing.

1. Zammit, V., Zammit, P., Savona-Ventura, C., Buhagiar, A. & Grech, V. Maltese national birth weight for gestational age centile values. *Malta Med. J.* **22,** 19–24 (2010).
2. NLS. at <http://www.resus.org.uk/pages/nls.pdf>
3. British Association of Perinatal Medicine. Position statement on early care of the newborn. *BAPM Publ.* at <http://www.bapm.org/publications/>

Appendix 4: Setting up a Neopuff® with air/oxygen blender

Setting up a Neopuff® with air/oxygen blender



1. Ensure that the air/oxygen blender is connected to air and oxygen supplies.
2. Connect the blender outlet to the Neopuff gas inlet as shown.
3. With the flow meter fully closed the manometer on the Neopuff should read zero.
4. Open the gas supply on the flow meter to 8L/min.
   Do not alter this flow rate during resuscitation

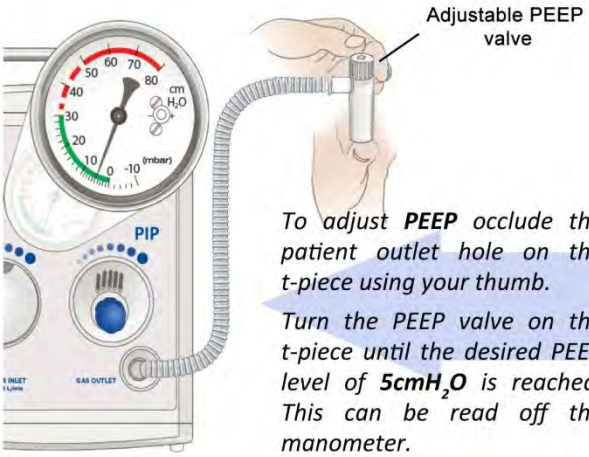
5. Adjust FiO_2 to the desired level on the air/oxygen blender.

   Start with 21% oxygen (room air) for term babies and 30% oxygen for preterm babies.

6. Connect the t-piece to the Neopuff gas outlet.
7. Adjust **PIP** and **PEEP** as shown below:

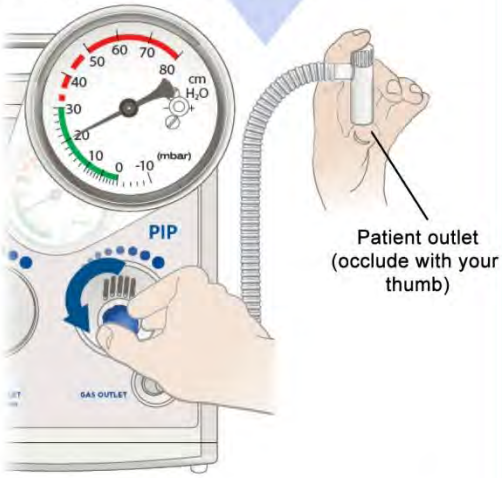
   *To adjust **PIP** occlude both outlets on thet-piece using your thumb (for the patient outlet) and your index finger (for the PEEP valve). Ensure that there are no leaks.*

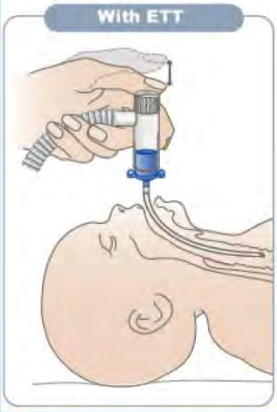
   Turn the PIP knob on the Neopuff unit until the desired PIP level is reached. This can be read off the manometer.



*To adjust **PEEP** occlude the patient outlet hole on the t-piece using your thumb.*

*Turn the PEEP valve on the t-piece until the desired PEEP level of **5cmH$_2$O** is reached. This can be read off the manometer.*





- The patient end of a t-piece can be attached to both a facemask and ETT.
- Starting PIP should be:
  30cmH$_2$O for term babies
  20-25cm H$_2$O for preterm babies
- PIP, PEEP and FiO_2 can be adjusted as needed during resuscitation.
- Inflations are delivered by occluding the PEEP valve for the desired duration.

newborn resuscitation guidelines 2013

HOSPITAL MATER DEI

Appendix 5: Central delivery suite daily checklist

SPTAR MATER DEI

Central Delivery Suite

Neonatal resuscitation room daily checklist

Equipment	Qty
Neonatal resuscitaires	3
Pulse oximeters	3
Neopuff	3
Neopuff circuit (T-piece)	3
Air/oxygen blender	3
Baby weighing scales	1
Chlorohexidine wipes packet	1
Aprons (roll)	1
Non-sterile gloves box S,M,L	1 each

Equipment on every resuscitaire shelf	
Suction apparatus	1
Stethoscope	1
Self-inflating (Ambu®) bag 450ml	1
Facemasks size 1 (large)	1
Facemasks size 0 (medium)	1
Facemasks size 00 (small)	1
Laryngoscope	1
Laryngoscope blade size 1	1
Laryngoscope blade size 0	1
Laryngoscope blade size 00	1
Suction catheters size 8, 10, 12	5 each
Yankauer sucker	1
Resuscitation chart	1

Equipment in every resuscitaire drawer	
Umbilical cord clamps	5
Surgical blades	5
Gauze swabs	5
Warm towels	**5**

Equipment in resuscitation trolley	
Spare laryngoscope with 3 blades	1
Uncuffed endotracheal tubes size 2.0	3
Uncuffed endotracheal tubes size 2.5	3
Uncuffed endotracheal tubes size 3.0	3
Uncuffed endotracheal tubes size 3.5	3
Uncuffed endotracheal tubes size 4.0	3
Endotracheal tube introducer (stylet) + fixers	3
Oropharyngeal airway size 000	3
Oropharyngeal airway size 00	3
Nasogastric tubes 4F, 6F	5 each
Umbilical venous catheter sets	3
Umbilical venous catheter size 6G, 4G	2 each
Silk 2/0 suture	3
Intraosseous needle 18G (pink)	3
Needles 21G (green)	10
Needles 25G (orange)	10
Cannula 24G (yellow)	10
Syringes 5ml	5
Syringes 10ml	5
Blood gas syringes (heparinised)	3
Butterfly needle (blue)	5
Three-way tap	3
Adhesive tape roll	2
0.9% saline ampoules 10ml	10
Prefilled adrenaline 1:10,000	3
Prefilled sodium bicarbonate 4.2%	3
Dextrose 10% 500ml bag	1
Naloxone hydrochloride 400mcg/ml ampoules	3
Pulse oximeter probes	5
Measuring tape	1
Polythene food grade bags roll	1
Sterile gloves sizes 6-8	5 each
Scissors	1

Daily check routine

- Check that pulse oximeter is functional
- Check air and oxygen supplies and suction at all points
- Set suction to 100mmHg
- Check laryngoscope batteries and ensure that bulb a tight fit
- Check contents of resuscitation room as per list above
- Ensure that towel warmer is switched on and that plenty of towels are available
- Keep both resuscitation room doors closed and a temperature of at least 25°C

Appendix 6: Transfer checklist

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Neonatal Paediatric Intensive Care Unit

In-hospital neonatal transfer checklist

Hardware	
Closed incubator	1
Multi-sensor monitor	1
Suction apparatus	1
Oxygen cylinder + flowmeter	1
Self-inflating (Ambu®) bag (500ml)	1
Facemasks size 1 (large)	1
Facemasks size 0 (medium)	1
Facemasks size 00 (small)	1
Laryngoscope	1
Laryngoscope blade size 1	1
Laryngoscope blade size 0	1
Laryngoscope blade size 00	1
Stethoscope	1
Measuring tape	1
Scissors	1

Pre-transfer check:

- Check that the battery on the closed incubator is fully charged. Keep plugged in when possible
- Check that closed incubator is pre-warmed to 33-35°C and pre-humidified to 80%

Incubator temperatures (at birth)	
GA weeks	Temp °C
< 30 weeks	35
30-35	34
>35	33

- If using a closed incubator with ventilator, check ventilator using test-lung
- Check that the multi-sensor monitor is functional
- Check that air/oxygen cylinders are full and that flow-meter works easily
- Check suction and set suction to 100mmHg
- Check laryngoscope and ensure that bulb is a tight fit
- If attending a delivery consider whether surfactant may be required. If so, the dose carried should be estimated according to the baby's gestational age

Disposables	
Chlorohexidine wipes packet	1
Aprons	5
Non-sterile gloves box S,M,L	1 each
Sterile gloves sizes 6-8	3
Uncuffed endotracheal tubes size 2.0	3
Uncuffed endotracheal tubes size 2.5	3
Uncuffed endotracheal tubes size 3.0	3
Uncuffed endotracheal tubes size 3.5	3
Uncuffed endotracheal tubes size 4.0	3
Endotracheal tube introducer (stylet)	3
ETT anchoring ties	3
Oropharyngeal airway size 000	3
Orophraryngeal airway size 00	3
Suction catheters size 5,8,10	5 each
Yankauer sucker	3
Nasogastric tubes 4F, 6F	5 each
Wool caps (bonnets)	assortment
Polythene food grade bags roll	1
Resuscitation chart	1
Pulse oximeter probes	5
Chest drain (black)	1
Flutter valve	2
Umbilical venous catheter sets	3
Umbilical venous catheter 6G, 4G	2 each
Silk 2/0 suture	2
Intraosseous needle 18G (pink)	2
Chlorhexidine wipes	10
Needles 21G (green)	10
Needles 25G (orange)	10
Cannula 24G (yellow)	10
Syringes 5ml	10
Syringes 10ml	5
Cannula fixers	5
Blood gas syringes (heparinised)	3
Butterfly needle (blue)	5
Three-way tap	3
Adhesive tape roll	1
Gauze swabs	5
Drugs	
0.9% saline ampoules 10ml	5
Prefilled adrenaline 1:10,000	3
Prefilled sodium bicarbonate 4.2%	2
Dextrose 10% 500ml bag	1
Naloxone 400mcg/ml ampoules	3
Suxamethonium ampoule 50mg/ml	2
Morphine ampoule 10mg/ml	2
Atropine ampoule 600mcg/ml	2

Surfactant – quantity depends on gestation of baby

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